IoT Homework 2025/2026

Exercise 1: 802.15.4

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1 Introduction

This report outlines the analytical design and performance evaluation of a wireless vibration monitoring system. The network is composed of one Personal Area Network (PAN) Coordinator and three ESP32-based vibration monitoring nodes. The communication protocol relies on the IEEE 802.15.4 standard operating in a beacon-enabled mode, restricted exclusively to the Collision Free Part (CFP) to ensure deterministic data delivery. Based on the system specifications, the nodes sample accelerometer data and transmit a payload every 10 seconds. The size of the payload is dynamically adjusted based on the number of vibration peaks detected within the reporting window. The following sections detail the statistical modeling of the traffic, the slot allocation strategy, the derivation of timing parameters, and a scalability analysis under strict duty cycle constraints.

2 Probability Mass Function of the Output Rate

The first step in analyzing the network traffic is to determine the Probability Mass Function (PMF) of the output bit rate for a single sensor node. According to the system setup, the number of vibration peaks detected in a 10-second window follows a Poisson distribution with an average rate of $\lambda = 0.4$ peaks per window.

The probability of observing exactly k peaks in a given interval is given by the Poisson formula:

$$P(k) = \frac{e^{-\lambda}\lambda^k}{k!} \tag{1}$$

By substituting $\lambda = 0.4$, we compute the probabilities for the three possible system states (0 peaks, 1 peak, and strictly more than 1 peak):

$$P(\text{peaks} = 0) = \frac{e^{-0.4} \cdot 0.4^0}{0!} \approx 0.6703 \tag{2}$$

$$P(\text{peaks} = 1) = \frac{e^{-0.4} \cdot 0.4^1}{1!} \approx 0.2681 \tag{3}$$

$$P(\text{peaks} > 1) = 1 - P(0) - P(1) \approx 1 - 0.6703 - 0.2681 = 0.0616 \tag{4}$$

The application logic dictates the payload size based on these states. Given the fixed reporting period (Beacon Interval, BI) of 10 seconds, we can calculate the corresponding data rates in bits per second (bps) for each state:

- **0 peaks (r_0):** Payload is 512 bytes.

$$r_0 = \frac{512 \text{ bytes} \times 8 \text{ bits/byte}}{10 \text{ s}} = 409.6 \text{ bps}$$

- **1 peak (r_1):** Payload is 2048 bytes (2 KB).

$$r_1 = \frac{2048 \text{ bytes} \times 8 \text{ bits/byte}}{10 \text{ s}} = 1638.4 \text{ bps}$$

- **$>$ 1 peak (r_2):** Payload is 4096 bytes (4 KB).

$$r_2 = \frac{4096 \text{ bytes} \times 8 \text{ bits/byte}}{10 \text{ s}} = 3276.8 \text{ bps}$$

Therefore, the exact PMF of the output rate is:

$$P(r) = \begin{cases} 0.6703, & \text{if } r = 409.6 \text{ bps} \\ 0.2681, & \text{if } r = 1638.4 \text{ bps} \\ 0.0616, & \text{if } r = 3276.8 \text{ bps} \end{cases} \tag{5}$$

3 Consistent CFP Slot Assignment

To achieve a "consistent" Guaranteed Time Slot (GTS) assignment within the CFP, the network must be dimensioned to prevent packet loss. While the statistical average of the required slots could be computed, allocating resources based on the average would lead to severe buffer overflows and packet drops whenever a node experiences a peak event (i.e., generating 2 KB or 4 KB payloads).

Thus, a robust network design requires a worst-case dimensioning approach. We are given the packet size $L = 128$ bytes and the condition that one packet perfectly fits into one slot. Let us calculate the number of slots required for each scenario:

- For r_0 (512 bytes): $\frac{512}{128} = 4$ slots
- For r_1 (2048 bytes): $\frac{2048}{128} = 16$ slots
- For r_2 (4096 bytes): $\frac{4096}{128} = 32$ slots

To guarantee null sample loss even during periods of high vibrational activity, each of the 3 monitoring nodes must be statically assigned the maximum required number of slots. Consequently, the consistent slot assignment is **32 slots per node**.

4 System Timing Parameters and Duty Cycle

With the slot allocation established, we can compute the precise timing parameters of the IEEE 802.15.4 superframe. The system operates with a nominal bit rate $R = 250$ kbps $(250,000$ bps).

First, we calculate the duration of a single slot (T_s):

$$T_s = \frac{L}{R} = \frac{128 \text{ bytes} \times 8 \text{ bits/byte}}{250,000 \text{ bps}} = \frac{1024 \text{ bits}}{250,000 \text{ bps}} = 0.004096 \text{ s} = 4.096 \text{ ms} \tag{6}$$

Next, we calculate the total number of slots required in the CFP ($N_{\text{slots}}^{\text{CFP}}$) for the 3 active nodes:

$$N_{\text{slots}}^{\text{CFP}} = 3 \text{ nodes} \times 32 \text{ slots/node} = 96 \text{ slots} \tag{7}$$

The active portion of the superframe (T_{active}) comprises the CFP slots plus exactly 1 slot dedicated to the beacon transmission:

$$T_{\text{active}} = (N_{\text{slots}}^{\text{CFP}} + 1) \times T_s = 97 \times 4.096 \text{ ms} = 397.312 \text{ ms} \tag{8}$$

Since the reporting period is 10 seconds, the Beacon Interval (BI) is exactly 10 seconds. The inactive portion of the superframe (T_{inactive}) is the time remaining after the active part completes:

$$T_{\text{inactive}} = BI - T_{\text{active}} = 10 \text{ s} - 0.397312 \text{ s} = 9.602688 \text{ s} \tag{9}$$

Finally, the duty cycle (η), which dictates the energy efficiency of the network, is the ratio of the active time to the total beacon interval:

$$\eta = \frac{T_{\text{active}}}{BI} = \frac{0.397312 \text{ s}}{10 \text{ s}} = 0.0397312 \approx 3.97\% \tag{10}$$

5 Scalability Analysis under Duty Cycle Constraints

The current configuration yields a highly efficient duty cycle of approximately 3.97%. To understand the scalability of the network, we evaluate how many additional nodes can be introduced without exceeding a maximum duty cycle threshold of 10%.

A 10% duty cycle over a 10-second Beacon Interval enforces an upper bound on the active time:

$$T_{\text{active}}^{\max} = \eta_{\max} \times BI = 0.10 \times 10 \text{ s} = 1 \text{ s} = 1000 \text{ ms} \tag{11}$$

We compute the maximum total number of active slots that can fit within this 1000 ms window:

$$N_{\text{slots}}^{\max} = \left\lfloor \frac{T_{\text{active}}^{\max}}{T_s} \right\rfloor = \left\lfloor \frac{1000 \text{ ms}}{4.096 \text{ ms}} \right\rfloor = 244 \text{ slots} \tag{12}$$

Subtracting the mandatory beacon slot, the maximum number of slots available for the CFP is:

$$N_{\text{CFP}}^{\max} = 244 - 1 = 243 \text{ slots} \tag{13}$$

Given the consistent assignment strategy of 32 slots per node, the absolute maximum number of nodes the network can support is:

$$\text{Total Nodes} = \left\lfloor \frac{243}{32} \right\rfloor = 7 \text{ nodes} \tag{14}$$

Since the network currently operates with 3 nodes, the system can safely accommodate **4 additional monitoring nodes** without violating the 10% duty cycle restriction.

6 Summary of Numerical Results

To provide a clear overview of the system's operational parameters derived in the previous sections, the key numerical findings are summarized in Table 1.

Parameter	**Value**
$P(r = r_0)$	0.6703
$P(r = r_1)$	0.2681
$P(r = r_2)$	0.0616
Assigned Slots per Node	32
T_s	4.096 ms
$N_{\text{slots}}^{\text{CFP}}$	96
T_{active}	397.312 ms
T_{inactive}	9.602688 s
Duty Cycle (η)	3.97%
Max Additional Nodes	4

Table 1: *Summary of calculated parameters for the IEEE 802.15.4 vibration monitoring network.*

7 Conclusion

The analysis demonstrates that worst-case dimensioning is a strictly necessary approach to guarantee deterministic delivery for sporadic high-load events (Poisson-distributed vibration peaks) in a beacon-enabled IEEE 802.15.4 network. Despite this conservative resource allocation, the current 3-node topology remains highly energy-efficient, maintaining a duty cycle of 3.97%. Furthermore, the network exhibits solid scalability, capable of more than doubling its capacity (up to 7 total nodes) while remaining safely under a strict 10% duty cycle operational limit.